SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I **V Thanisha / 1965260160** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of student: V Thanisha _____

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A
Debugging & Optimisation Task

Question 1

A network administrator has configured a workstation with the following settings:
IP Address: 192.168.10.65
Subnet Mask: 255.255.255.192 (/26)
Default Gateway: 192.168.10.1
However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:
• Identifying the subnet to which the configured IP address belongs.
• Determining the network address, broadcast address, and valid host range.
• Verifying whether the configured IP address and gateway are valid.
• Identifying the configuration errors.
• Optimizing the configuration by assigning an appropriate IP address and default gateway.
• Calculating the total number of usable host addresses after correction.

Answer

Step 1 - Given: IP 192.168.10.65, mask /26, gateway 192.168.10.1.

Step 2 - Calculate block size: 256 - 192 = 64. The /26 subnets are .0-.63, .64-.127, .128-.191, and .192-.255.

Step 3 - Current subnet: 192.168.10.65 belongs to 192.168.10.64/26. Network = 192.168.10.64; broadcast = 192.168.10.127; valid hosts = 192.168.10.65 through 192.168.10.126.

Step 4 - Find error: The configured IP is a valid host of the .64/26 subnet, but the gateway 192.168.10.1 is in the different .0/26 subnet. The workstation is also not in the intended 192.168.10.0/26 subnet.

Step 5 - Correct configuration: For the intended .0/26 subnet, assign IP 192.168.10.2, mask 255.255.255.192 and gateway 192.168.10.1, provided that gateway is configured on the router and the address is unused.

Step 6 - Check corrected range: Network 192.168.10.0; broadcast 192.168.10.63; valid hosts .1-.62. Both .1 and .2 belong to the same subnet.

Step 7 - Usable hosts: Host bits = 32 - 26 = 6. Usable hosts = 2^6 - 2 = **62**.

Result: The mismatch is corrected by placing the workstation and gateway in 192.168.10.0/26; verify connectivity with ping.

Question 2

Two devices are configured with the following IPv6 addresses:
Device A: 2001:db8::ff00:42:8329/64
Device B: 2001:db8:0:0:1::1/64
Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:
• Expanding both IPv6 addresses into their complete notation.
• Determining the network prefix of each device.
• Verifying whether both devices belong to the same subnet.
• Identifying the addressing or prefix mismatch.
• Optimizing the IPv6 configuration by assigning appropriate addresses.
• Calculating the number of available host addresses within the corrected subnet.

Answer

Step 1 - Expand Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329/64.

Step 2 - Expand Device B: 2001:0db8:0000:0000:0001:0000:0000:0001/64.

Step 3 - Identify /64 prefix: The first 64 bits are the first four groups. For both devices these are 2001:0db8:0000:0000.

Step 4 - Compare networks: Both have network prefix **2001:db8:0:0::/64**. The fifth group differs but is in the interface identifier, not the /64 prefix.

Step 5 - Diagnose: There is **no IPv6 prefix mismatch** in the provided addresses. The reason for failed communication cannot be determined from the addresses alone. Check interface status, VLANs, neighbor discovery, firewall and duplicate addresses.

Step 6 - Optimization: Keep the existing distinct addresses, or choose simpler addresses such as 2001:db8::1/64 and 2001:db8::2/64 on the same link. Check with IPv6 ping and neighbor tables.

Step 7 - Address count: Interface bits = 128 - 64 = 64; possible interface identifiers = **2^64 = 18,446,744,073,709,551,616**. Some identifiers have special uses; this is the theoretical count.

Result: Both supplied devices are already in the same /64; investigate non-address causes.

Question 3

An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:
• Identifying the network address, broadcast address, and host range for each subnet.
• Calculating the number of usable and unused IP addresses.
• Identifying inefficient subnet allocations.
• Debugging the addressing plan by locating subnet allocation mistakes.
• Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

Answer

Step 1 - Given: Parent network 192.168.1.0/24 (256 total addresses). Department counts and existing assigned subnets are not provided, so an exact utilization audit is not possible.

Step 2 - Capacities: /26 has 64 total, 62 usable hosts; /27 has 32 total, 30 usable; /28 has 16 total, 14 usable. Formula: usable IPv4 hosts = 2^(32 - prefix) - 2.

Step 3 - Example non-overlapping allocation: Department A: 192.168.1.0/26, broadcast .63, hosts .1-.62; Department B: 192.168.1.64/27, broadcast .95, hosts .65-.94; Department C: 192.168.1.96/28, broadcast .111, hosts .97-.110.

Step 4 - Count allocated and free: The example uses 64 + 32 + 16 = 112 addresses; 256 - 112 = **144 unallocated addresses** (.112-.255). Its three subnets support 62 + 30 + 14 = **106 usable hosts**.

Step 5 - Unused host addresses: If A, B, C require a, b, c host IPs, unused usable addresses = (62 - a) + (30 - b) + (14 - c), assuming requirements fit. Actual values cannot be calculated without device counts.

Step 6 - Debugging: Check that subnet starts align with their block sizes; ensure no subnet overlaps another; exclude network/broadcast addresses; ensure each gateway and host use the correct mask.

Step 7 - VLSM method: Sort departments by required hosts (largest first). Select the smallest block where 2^h - 2 is at least the host requirement; assign contiguous, correctly aligned blocks from the beginning of the /24.

Step 8 - Optimization: Allocate /26 only where up to 62 hosts are needed, /27 for up to 30, and /28 for up to 14. Smaller needs may use /29 (6 usable) when appropriate.

Result: The sample allocation is valid and leaves 144 unallocated addresses. A final optimal plan needs each department's actual host count.

Question 4

Two devices are configured with the following IPv6 addresses:
Device A: 2001:db8::ff00:42:8329/64
Device B: 2001:db8:0:0:1::1/64
Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:
• Expanding both IPv6 addresses into their complete notation.
• Determining the network prefix of each device.
• Verifying whether both devices belong to the same subnet.
• Identifying the addressing or prefix mismatch.
• Optimizing the IPv6 configuration by assigning appropriate addresses.
• Calculating the number of available host addresses within the corrected subnet.

Answer

Step 1 - Address expansion: Device A = 2001:0db8:0000:0000:0000:ff00:0042:8329; Device B = 2001:0db8:0000:0000:0001:0000:0000:0001.

Step 2 - Prefix: /64 means the first four 16-bit groups identify the network. Both prefixes are 2001:0db8:0000:0000::/64.

Step 3 - Subnet verification: Both devices are in **2001:db8::/64**, so the addresses alone do not explain the failure.

Step 4 - Troubleshoot: Verify link status and same VLAN, IPv6 interface configuration, neighbor discovery, duplicate address detection and firewall ICMPv6 rules.

Step 5 - Suggested addresses: Device A 2001:db8::1/64 and Device B 2001:db8::2/64 are simpler valid examples on the same subnet. No change is required solely to fix a prefix mismatch.

Step 6 - Test: Check IPv6 neighbor entries and send an IPv6 ping to the peer; if unsuccessful, inspect link-layer and filtering settings.

Step 7 - Capacity: 128 - 64 = 64 interface bits; theoretical addresses = 2^64 = **18,446,744,073,709,551,616**.

Result: The supplied prefixes match; the actual communication fault needs additional network checks.

Question 5

A router contains the following routing table:
Destination Network — Next Hop
192.168.1.0/24 — 10.0.0.2
192.168.2.0/24 — 10.0.0.3
Default Route — 10.0.0.1
Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:
• Identifying the matching routing entry using the longest-prefix match.
• Determining the next-hop router selected for packet forwarding.
• Verifying whether the destination IP belongs to the intended subnet.
• Identifying any routing configuration errors.
• Optimizing the routing table to improve packet forwarding efficiency.
• Identifying the route that will be used if no matching destination exists.

Answer

Step 1 - Destination: 192.168.2.100. Compare it with each routing table prefix.

Step 2 - Longest-prefix match: 192.168.2.100 matches 192.168.2.0/24 (255.255.255.0). It does not match 192.168.1.0/24. The default /0 matches only as a fallback.

Step 3 - Selected next hop: Router forwards traffic for 192.168.2.100 to **10.0.0.3**.

Step 4 - Verify host range: Network 192.168.2.0, broadcast 192.168.2.255, valid hosts 192.168.2.1-.254; 192.168.2.100 is a valid host address.

Step 5 - Find possible error: The given routing table contains no obvious prefix-selection error. Check whether 10.0.0.3 is reachable, whether that router has a path to 192.168.2.0/24, and whether a return route exists.

Step 6 - Debugging checks: Inspect routing and interface state; ping 10.0.0.3; use traceroute toward 192.168.2.100; check access-control lists and destination host connectivity.

Step 7 - Optimize: Keep the specific /24 route when correct. If 10.0.0.3 is wrong, replace it with the verified next hop; remove duplicate or stale routes and ensure reverse-path routing.

Step 8 - No match: Destinations not matching a more specific entry use the default route via **10.0.0.1**.

Result: Expected route is 192.168.2.0/24 via 10.0.0.3; investigate reachability or downstream routing.